whereas $c_\kappa \ll c_u$ in the soft-core case. This emphasizes the contrasting importance of compressibility and lattice elasticity in the properties of the two supersolids.

1. Bulk incompressible limit

We take the bulk incompressible limit to be when $c_u \ll c_\kappa$, which is relevant to the dipolar supersolid. Within this regime c_t is unaffected but c_1 and c_2 become

$$c_1 \to \sqrt{c_\kappa^2 + f_n c_u^2}, \tag{18}$$

$$c_2 \to \sqrt{f_s} c_u, \tag{19}$$

where $f_n = 1 - f_s$ is the normal fraction. These two limits approximately describe the first and second sound velocity in the dipolar model, as shown in Fig. 6(a).

2. Rigid lattice limit

The rigid lattice limit occurs when $c_u \gg c_\kappa$. As noted before, this is the regime appropriate to the soft-core supersolid. Within this regime c_t is unaffected but c_1 and c_2 become

$$c_1 \to \sqrt{c_u^2 + f_n c_\kappa^2}, \tag{20}$$

$$c_2 \to \sqrt{f_s} c_\kappa. \tag{21}$$

These two limits approximately describe the first and second sound velocity in the soft-core 2D model, as shown in Fig. 6(b).

The extreme case $c_u \to \infty$ can be realised with a BEC loaded into an optical lattice[6]. Here the lattice sites cannot move (cf. Refs. [48, 49]) and the system has a single gapless band corresponding to second sound.

V. OUTLOOK AND CONCLUSIONS

In this paper we have contrasted the excitation spectra of $D = 2$ supersolids arising in two different systems. We have focused on the three gapless excitation branches and their associated speeds of sound. Our work demonstrates that hydrodynamic theory provides an accurate prediction of the speeds of sound based only on a set of generalized elastic parameters that are determined from ground state calculations. We find that the dipolar system is dominated by its compressibility ($c_\kappa \gg c_u$) whereas the soft-core system is in the rigid lattice regime ($c_\kappa \ll c_u$). As a result the first sound speed of the dipolar system is significantly larger than the other speeds of sound, and there is barely any jump in the first sound speed at the superfluid to supersolid transition. A recent proposal has suggested a method for measuring the long-wavelength excitation frequencies, and longitudinal speeds of sound in a 1D supersolid using a periodic optical potential [15]. This method can be directly extended to the 2D supersolid.

Our results will provide a basis for better understanding the equilibrium and dynamical properties of higher dimensional supersolids, which are being explored in experiments with dipolar gases. We have also calculated the excitations and compared to the hydrodynamic theory for a higher density dipolar BEC ($\rho = 0.08/a_{dd}^2$) and find results that are qualitatively similar to those presented here, except that c_κ is slightly larger. An interesting remaining question is to extend our analysis to even higher densities, where the dipolar supersolid transitions to a stripe state or a hexagonal state. An interesting aspect of the stripe state that its superfluid fraction and elastic tensor will become anisotropic. Such high densities will be difficult to reach in experiments with magnetic gases, but these parts of the phase diagram might be accessible to polar molecules gases which have strong dipole-dipole interactions [50, 51]. This presents a rich playground for studying superfluidity in the presence of crystalline order.

Note added. Recently the preprint Ref. [52] appeared, which also considers the elastic properties of soft-core supersolids.

ACKNOWLEDGMENTS

DB and PBB acknowledge use of New Zealand eScience Infrastructure (NeSI) high performance computing facilities and support from the Marsden Fund of the Royal Society of New Zealand. We acknowledge useful conversations and comparisons with P. Senarath Yapa and T. Bland. E.P. acknowledges support by the Austrian Science Fund (FWF) within the DK-ALM (No. W1259-N27).

Appendix A: Planar dipolar BEC

Here we discuss the details of our model of a planar dipolar BEC of magnetic atoms. The atoms are free move in the xy-plane and the planar kinetic energy is given by

$$T_{\mathbf{v}} = \frac{(-i\hbar\boldsymbol{\nabla}_\perp + m\mathbf{v})^2}{2m}, \tag{A1}$$

where $\boldsymbol{\nabla}_\perp$ is the 2D gradient and we have allowed for a superfluid velocity $\mathbf{v}$ (cf. [8, 53]). In the z-direction the single particle Hamiltonian includes harmonic confinement

$$H_z = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial z^2} + \frac{1}{2}m\omega_z^2 z^2, \tag{A2}$$

with ω_z being the angular frequency. The magnetic dipole moments of the atoms are taken to be polarized along z by a bias field and the interactions are described by the potential

$$U(\mathbf{r}) = \frac{4\pi a_s \hbar^2}{m}\delta(\mathbf{r}) + \frac{3a_{dd}\hbar^2}{mr^3}\left(1 - 3\frac{z^2}{r^2}\right), \tag{A3}$$

[6] This is an artificial supersolid, because translational invariance is not spontaneously broken.

where $\mathbf{r} = \mathbf{x} - \mathbf{x}'$ is the relative separation between the particles. Here a_s is the$s$-wave scattering length, $a_{dd} = m\mu_0\mu_m^2/12\pi\hbar^2$ is the dipole length, and μ_m is the atomic magnetic moment. The ratio $\epsilon_{dd} = a_{dd}/a_s$ characterises the relative strength of the dipole-dipole to s-wave interactions and when this parameter is sufficiently large the ground state undergoes a transition to a crystalline state with spatial modulation in the xy-plane.

The eGPE energy functional for this system is

$$E = \int_{\rm uc} d\mathbf{x}\, \psi^* \left(T_{\mathbf{v}} + H_z + \tfrac{1}{2}\Phi + \tfrac{2}{5}\gamma_{\rm QF}|\psi|^3\right)\psi, \tag{A4}$$

where we have introduced the effective potential

$$\Phi(\mathbf{x}) = \int d\mathbf{x}'\, U(\mathbf{x}-\mathbf{x}')|\psi(\mathbf{x}')|^2, \tag{A5}$$

and the effects of quantum fluctuations are described by the term with coefficient $\gamma_{\rm QF} = \frac{128\pi\hbar^2}{3m}a_s\sqrt{\frac{a_s^3}{\pi}}\mathcal{Q}_5(\epsilon_{dd})$, where $\mathcal{Q}_5(x) = \Re\{\int_0^1 du[1 + x(3u^2-1)]^{5/2}\}$ [54]. Because the system is infinite in the xy-plane we restrict the spatial extent of the wavefunction to the unit cell (uc) defined by the direct lattice vectors $\{\mathbf{a}_1, \mathbf{a}_2\}$ and impose periodic boundary conditions. To accurately calculate $\Phi(\mathbf{x})$ in the unit cell we employ the z-cutoff truncated interaction potential introduced in Ref. [55].

The average density condition enforces the following normalization constrain on the wavefunction

$$\int_{\rm uc} d\mathbf{x}\, |\psi(\mathbf{x})|^2 = \rho A, \tag{A6}$$

where $A = |\mathbf{a}_1 \times \mathbf{a}_2|$ is the area of the unit cell. The ground state is determined for a specified value of ρ and $\mathbf{v} = \mathbf{0}$ by minimising the energy density

$$\mathcal{E}(\rho, \mathbf{a}_1, \mathbf{a}_2, \mathbf{v}) \equiv E/A. \tag{A7}$$

with respect to the unit cell parameters $\{\mathbf{a}_1, \mathbf{a}_2\}$. For the purpose of computing the elastic parameters, we are also interested in small changes in the parameters $\{\rho, \mathbf{a}_1, \mathbf{a}_2, \mathbf{v}\}$ (i.e. to make finite difference derivatives) from the ground state values. In this case the energy density is minimised with respect to ψ, but with the other parameters specified.

Appendix B: 2D soft-core condensate

The 2D soft-core model is for a BEC of atoms in the xy-plane and interacting with a potential of the form $U_{\rm sc}(\mathbf{r}) = U_0\theta(a_{\rm sc} - |\mathbf{r}|)$, where $a_{\rm sc}$ is the soft-core radius, U_0 is the potential strength, and θ is the Heaviside step function. The meanfield energy functional for this system is similar to the dipolar case, but with out any z-direction or quantum fluctuations

$$E = \int_{\rm uc} d\mathbf{x}\, \psi^* \left(T_{\mathbf{v}} + \tfrac{1}{2}\Phi_{\rm sc}\right)\psi, \tag{B1}$$

where

$$\Phi_{\rm sc}(\mathbf{x}) = \int d\mathbf{x}'\, U_{\rm sc}(\mathbf{x}-\mathbf{x}')|\psi(\mathbf{x}')|^2. \tag{B2}$$

In this system it is convenient to adopt $a_{\rm sc}$ as the unit of length and $\hbar\omega_0 = \hbar^2/ma_{\rm sc}^2$ as the unit of energy, and to define the dimensionless interaction parameter

$$\Lambda = \frac{m\pi\rho a_{\rm sc}^4 U_0}{\hbar^2}. \tag{B3}$$

The normalization condition, definition of energy density, and procedure to obtain ground states is then the same as described for the dipolar case.

It is worth noting that as discussed by Macrì *et al.* [28], for any finite value of Λ the meanfield description of 2D soft-core system will be valid for sufficiently high density ρ. There have also been comparisons validating the meanfield 2D soft-core model against quantum Monte Carlo results in Ref. [28] (also see [37]).

Appendix C: Bogoliubov-de Gennes equations

The Bogoliubov-de Gennes (BdG) framework describes the quasiparticle excitations about a stationary solution[7] $\psi(\mathbf{x})$. This solution satisfies the time-independent GPE $\mathcal{L}\psi = \mu\psi$, where

$$\mathcal{L} = T_{\mathbf{0}} + \Phi_{\rm sc}, \qquad \text{(soft-core)} \tag{C1}$$

$$\mathcal{L} = T_{\mathbf{0}} + H_z + \Phi + \gamma_{\rm QF}\psi^3, \qquad \text{(dipolar)} \tag{C2}$$

and μ is the chemical potential.

The quasiparticle modes can be determined by linearising the time-dependent evolution, $i\hbar\dot{\Psi} = \mathcal{L}\Psi$, with an expansion of the form given in Eq. (2). This leads to the BdG equations

$$\begin{pmatrix} \mathcal{L} + X - \mu & -X \\ X & -(\mathcal{L} + X - \mu) \end{pmatrix} \begin{pmatrix} u_{\nu\mathbf{q}} \\ v_{\nu\mathbf{q}} \end{pmatrix} = \hbar\omega_{\nu\mathbf{q}} \begin{pmatrix} u_{\nu\mathbf{q}} \\ v_{\nu\mathbf{q}} \end{pmatrix}, \tag{C3}$$

where $u_{\nu\mathbf{q}}$ and $v_{\nu\mathbf{q}}$ have the Bloch form $f_{\nu\mathbf{q}}(\mathbf{x}) = \bar{f}_{\nu\mathbf{q}}(\mathbf{x})e^{i\mathbf{q}\cdot\mathbf{x}}$ with $\bar{f}_{\nu\mathbf{q}}(\mathbf{x})$ periodic in the unit cell, and X is defined so that

$$Xf = \psi\int d\mathbf{x}' U_{\rm sc}(\mathbf{x}-\mathbf{x}')f(\mathbf{x}')\psi(\mathbf{x}'), \qquad \text{(soft-core)} \tag{C4}$$

$$Xf = \psi\int d\mathbf{x}' U(\mathbf{x}-\mathbf{x}')f(\mathbf{x}')\psi(\mathbf{x}')\ldots + \frac{3}{2}\gamma_{\rm QF}\psi^3 f. \qquad \text{(dipolar)} \tag{C5}$$

[7] Here we consider solutions for $\mathbf{v} = 0$ where ψ can be taken to be real.

Appendix D: Euler Lagrange equations

From Eq. (10) we obtain the Euler-Lagrange equations to describe the evolution of the hydrodynamic fields

$$\hbar\partial_t\phi = -\alpha_{\rho\rho}\delta\rho - \alpha_{\rho u}\partial_i u_i, \tag{D1}$$

$$m(\partial_t\delta\rho - \rho_n\partial_{ti}u_i) = -\hbar\rho_s\partial_i^2\phi, \tag{D2}$$

$$\rho_n(m\partial_t^2 u_i - \hbar\partial_{ti}\phi) = \alpha_{\rho u}\partial_i\delta\rho + C_{ijkl}\partial_{jk}u_l. \tag{D3}$$

We can further decompose $\mathbf{u} = \mathbf{u}_t + \mathbf{u}_l$ in to transverse ($\mathbf{u}_t$ with $\nabla\cdot\mathbf{u}_t = 0$) and longitudinal ($\mathbf{u}_l$ with $\nabla\times\mathbf{u}_l = 0$) parts. Using this decomposition and Eq. (8), the last Euler-Lagrange equation can be written as

$$\rho_n(m\partial_t^2\mathbf{u}_l - \hbar\partial_t\nabla\phi) = \alpha_{\rho u}\nabla\delta\rho + \alpha_{uu}\nabla^2\mathbf{u}_l, \tag{D4}$$

$$\rho_n m\partial_t^2\mathbf{u}_t = \tilde{\mu}\nabla^2\mathbf{u}_t. \tag{D5}$$

The normal mode solutions of these equations are of the form $X = X_w e^{i(\mathbf{q}\cdot\mathbf{x}-\omega t)}$, where $X = \{\delta\rho, \phi, u_l, u_t\}$ and X_w is the excitation amplitude (also see [14, 47]). Three non-trivial solutions can be found with dispersion relations of the form $\omega = cq$, yielding the speeds of sound given in Eqs. (11)-(13).

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